Real-time Whole-body Model Predictive Control for Humanoid Locomotion with Novel Kino-dynamic Model and Warm-start Method

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Abstract: Advancements in optimization solvers and computing power have led to growing interest in applying whole-body model predictive control (WB-MPC) to bipedal robots. However, the high degrees of freedom and inherent model complexity of bipedal robots pose significant challenges in achieving fast and stable control cycles for real-time performance. This paper introduces a novel kino-dynamic model and warm-start strategy for real-time WB-MPC in bipedal robots. Our proposed kino-dynamic model combines the linear inverted pendulum plus flywheel and full-body kinematics model. Unlike the conventional whole-body model that rely on the concept of contact wrenches, our model utilizes the zero-moment point (ZMP), reducing baseline computational costs and ensuring consistently low latency during contact state transitions. Additionally, a modularized multi-layer perceptron (MLP) based warm-start strategy is proposed, leveraging a lightweight neural network to provide a good initial guess for each control cycle. Furthermore, we present a ZMP-based whole-body controller (WBC) that extends the existing WBC for explicitly controlling impulses and ZMP, integrating it into the real-time WB-MPC framework. Simulations and real robot experiments further validate that the proposed framework demonstrates robustness to perturbation and satisfies real-time control requirements during walking.

Keywords: Bipedal robot, bipedal locomotion, whole-body model predictive control, optimal control

1. INTRODUCTION

Model predictive control (MPC) has significantly advanced walking robot technology by enabling stable and adaptive locomotion across various terrains. Based on a dynamics model, The MPC scheme predicts future states and optimizes control inputs, allowing bipedal robots to respond effectively against perturbations.

In this study, we propose a novel kino-dynamic model and warm-start strategy to reduce the computational burden and enable the real-time WB-MPC. The proposed model combines LIPFM and FKM, utilizing ZMP instead of contact wrenches, distinguishing it from other WB models. This approach allows for the consideration of dynamic effects of WB motion without requiring inverse dynamics calculations and avoids the discontinuities in the formulation of WB-MPC problem caused by changes in the contact phase.

Additionally, our proposed warm-start strategy utilizes a modularized and lightweight multi-layer perceptron (MLP), unlike existing machine learning-based memory of motion strategy. By modularizing and lightweighting the prediction model, our approach achieves a significantly lower latency in finding a good initial guess compared to previous studies. This enables rapid solving of DDP at each control cycle. All of these ensure real-time control performance and robust stability against perturbation. To implement MPC effectively, a whole-body controller (WBC) is crucial, as demonstrated in [1] and [2]. In this work, we improve the robustness of our WB-MPC framework by introducing a ZMP-based WBC which controls both the ZMP and impulses during walking.

2. PROPOSED WB-MPC FRAMEWORK

2.1 Novel Kino-dynamic Model

Previous studies proposed models for reflecting the WB motion, such as WBD or centroidal dynamics with FKM. However, these models have high complexity and cause problem discontinuities during contact state transitions, hindering real-time control performance [**?**]. To address these issues, we propose a novel kino-dynamic model that combines LIPFM and FKM. The proposed model eliminates problem discontinuities during contact state transitions and has relatively low model complexity by replacing the concept of contact wrenches with ZMP.

The LIPFM is the simplest ZMP-based model capable of reflecting angular momentum, without causing problem discontinuities during contact state transitions. The equations of the LIPFM are as follows.

$$\ddot{x} = \frac{g}{z_c}(x - p_x) - \frac{\tau_y}{m z_c} \tag{1}$$

$$\ddot{y} = \frac{g}{z_c}(y - p_y) + \frac{\tau_x}{m z_c} \tag{2}$$

where, x and y are the center of mass (COM) coordinates in the x and y-direction, p_x and p_y are the ZMP coordinates in the x and y-direction, τ_x and τ_y are the time derivative of centroidal angular momentum (CAM) in the x and y-direction, $\ddot{x}$ and $\ddot{y}$ are the accelerations of the COM in the x and y-direction, g is the gravity force, z_c is the height of the COM, and m is the total mass of the robot, respectively.

The time derivative of CAM generated by WB motion can be expressed from the centroidal momentum matrix as follows.

$$\boldsymbol{H}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \dot{\boldsymbol{H}}(\boldsymbol{q})\,\dot{\boldsymbol{q}} = \boldsymbol{\tau} \tag{3}$$

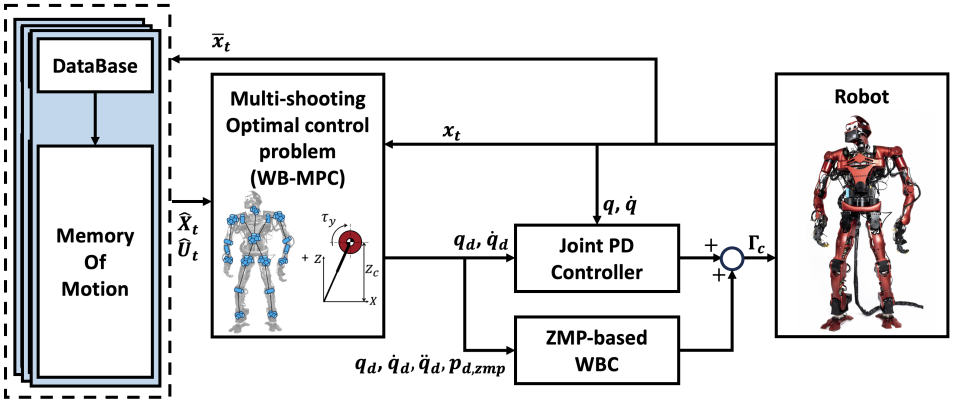


Fig. 1. The proposed control framework using novel kino-dynamic model.

where, $\boldsymbol{H}(\boldsymbol{q})$ and $\dot{\boldsymbol{H}}(\boldsymbol{q})$ are the angular part of centroidal momentum matrix and its time derivative, $\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$ are the velocity and acceleration of joints, and $\boldsymbol{\tau}$ is the time derivative of CAM, respectively.

Also, kinematic constraints for WB motion can be given through a FKM. The constraints of the COM and placement of end-effectors are determined as follows.

$$\boldsymbol{T}_e = \bar{X}_e(\boldsymbol{q}) \tag{4}$$

$$\boldsymbol{x}_{com} = \bar{X}_{com}(\boldsymbol{q}) \tag{5}$$

where, $\boldsymbol{T}_e$ is the placement of end-effectors and $\bar{X}_c$ is forward kinematics of end-effectors, $\boldsymbol{x}_{com}$ is the position of COM and $\bar{X}_{com}$ is forward kinematics of COM, respectively.

2.2 Proposed WB-MPC Framework

As illustrated in Fig. 1, the proposed WB-MPC framework not only utilizes the newly proposed kino-dynamic model for WB-MPC but also includes an MLP-based warm-starting algorithm that computes a good initial guess at each control cycle to solve WB-MPC quickly, as well as a WBC that regulates the zero-moment point (ZMP) and impact forces generated during contact.

3. SIMULATION AND REAL ROBOT EXPERIMENT

The WB-MPC used in the experiments employed a 60-step prediction horizon, with each step corresponding to a 0.02 s interval. As a result, the proposed control framework demonstrated performance of warm-starting algorithm and showed greater robustness to external disturbances during flat-ground walking compared to state-of-the-art algorithms based on other whole-body or simplified models. This was also implemented on a real robot, where WB-MPC adjusted the COM, ZMP, and CAM trajectories to maintain balance during external disturbances, and consequently, the robot's upper-body joints were utilized.

4. CONCLUSION

This paper presents a real-time WB-MPC framework for bipedal robot locomotion, featuring a novel kino-dynamic model and an efficient warm-start strategy. The proposed kino-dynamic model combines the a ZMP-based dynamics model with FKM, significantly reducing computational burden and resolve peak latency issues during contact state transitions compared to the previous WB model. This ensures the real-time performance of WB-MPC across all walking phases. Furthermore, the proposed modularized MLP based warm-start strategy demonstrates superior performance over existing memory of motion algorithms by providing a good initial guess at every control cycle, significantly leveraging real-time applicability. We also introduced a ZMP-based WBC, specifically adapted to complement for our WB-MPC framework. Through various experiments, the computational efficiency and robustness of our WB-MPC framework have been validated. Even under perturbation, we can solve the WB-MPC with 60 time steps, 4620 variables, 2400 constraints, and various penalty-form constraints within a latency of 17 ms, while the robot maintains balance.

However, the proposed kino-dynamic model, based on the ZMP concept with vertical COM constraints, has limitations in ensuring stability on terrains requiring vertical COM motion, such as stairs or steep slopes, and cannot account for the flight phase, limiting its ability to generate dynamic motions like running or jumping. Future work will focus on extending the model to overcome these limitations.

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